Milestone #4: System and Evaluation

1. *Evaluation Design*

Methods*:* We evaluated our prototype using two complementary methods; a heuristic evaluation and user testing sessions. For the heuristic evaluation, two students who were not involved in the design independently reviewed the interface using Nielsen's 10 usability heuristics. They assessed how well the system adhered to each heuristic and assigned severity ratings from 0 (not a problem) to 4 (catastrophic problem). This method enabled rapid identification of usability issues based on established design principles without requiring end users. In addition, we conducted user testing with five participants performing four representative tasks using the prototype on a mobile device inside PFT. This allowed us to observe real user behavior in a realistic environment.

Participants: We recruited five people who are familiar with Patrick F. Taylor Hall. We recruited the participants through word of mouth and by asking people in the building. Participants volunteered and were not compensated.

Table 1: Participants Demography

ID	Age	Gender	Role	Frequency in PFT	Tech Comfort
P1	22	Female	Undergraduate Student	3-4 times/week	Low
P2	28	Male	Graduate Student	Daily	Very High
P3	38	Female	Graduate Student	Daily	Medium
P4	24	Female	Graduate Student	2-3 times/week	Low
P5	31	Male	Graduate Student	Weekly	High

We recruited users with different levels of PFT experience and tech-savviness. P1 and P4 are novice users who struggle to navigate inside the building. P2 and P5 are power users who are efficient and see issues. P3 is a teaching assistant who could use the system to assist others, especially new students. This diversity allows us to check if the prototype is suitable for all types of users.

Procedure: Each session lasted approximately 25–30 minutes and was conducted at the entrance of PFT using a mobile device. Participants were first given a brief overview of the study and provided verbal consent. They were then asked to complete the following tasks: (i) Search for a destination (e.g., Room 1236 or a location such as Panera). (ii) Confirm the selected destination. (iii) Follow the displayed route on the map on the same floor. (iv) Find a route to a destination on another floor and interpret the floor transition instructions. Participants were also asked to try the custom starting location feature to find a route to another destination. During the tasks, participants were encouraged to use a think-aloud protocol. After each task, they rated the difficulty on a scale of 1–5 and described any confusion. At the end of the session, participants were asked about the most confusing part, whether they would use the system, and one suggested improvement.

Data Collection: We gathered three measures: (i) Time to complete tasks – Stopwatch. (ii) Observed usability issues - Measured by an observer noting participant clicks, verbalizations, and interface-related mistakes during the tasks. (iii) Perceived difficulty - 1-5 rating after each task. We recorded the sessions with participants' consent.

Analysis: We collected three types of data: task completion time (measured with a stopwatch), observed usability issues, and perceived difficulty ratings (1–5 scale). Sessions were recorded with participant consent. Quantitative data was analysed using average values. Qualitative data, including participant

comments and observed behaviors, was analysed using thematic analysis. One researcher identified recurring themes, and a second researcher reviewed them to ensure consistency.

2. Results: Main Findings:

Quantitative Results:

The quantitative analysis was aimed at the time spent on completing the tasks and the perceived ratings of the difficulties among the five participants. Instead of reporting the averages only, Table 2 provides the data of individual participants to show the high degree of variation in navigation performance.

Table 2 Task Results

Participant	Task 1 Time (sec)	Task 2 Time (sec)	Task 3 Time (min:sec)	Task 4 Time (min:sec)
P1	18	12	8:15	10:30
P2	12	8	6:45	8:20
P3	25	15	8:50	11:15
P4	20	10	7:30	9:05
P5	15	9	6:10	7:30

Tasks 1 and 2 (search and confirmation) had little individual difference with all participants taking less than 30 seconds to complete the tasks. Navigation tasks, though, showed a significant difference in users. The time spread of 2 minutes and 40 seconds, with a range of 6:10 (P5) to 8:50 (P3), was the range of values of task 3 (single-floor route following). Task 4 (cross-floor navigation) exhibited even more variation, with a range of 3 minutes and 45 seconds (between 7:30 (P5) and 11:15 (P3)).

The highest technology comfort (P2) and daily PFT familiarity were very high. Only medium technology comfort was noted by the slowest participant (P3). P5, who had weekly visits with PFT, completed cross-floor navigation in the shortest time (7:30), indicating that a general digital navigation skill can offset the absence of familiarity with the building. These differences in individuals show that although the prototype may be very effective with some users, it poses disproportionate challenges to other users, especially those who are less comfortable with technology. We intend to carry out a strict quantitative study with a bigger sample in the future, baseline comparison with the current methods of navigation (e.g., using existing ways of navigation), and paired t-tests to statistically justify the observed differences.

Qualitative Results: Thematic analysis of participant feedback revealed four major themes that characterize users' experiences with the prototype.

Theme 1: Efficient and Flexible Destination Search: Participants were able to locate and confirm destinations quickly and with minimal effort. All participants successfully completed the search and confirmation tasks without errors, indicating that the interface was intuitive and easy to use. Users appreciated the flexibility of the search feature, which supported both room numbers and point-of-interest names. One participant (P1) noted, *"I like that I can type 'restroom' instead of a room number. I don't usually remember exact numbers."* Another participant (P3) highlighted the benefit of flexible starting points, stating, *"Being able to pick where I'm starting from is helpful. I'm not always at the*

entrance." The destination card also improved user confidence by clearly displaying the selected location. These responses suggest that the system supports efficient destination lookup and reduces user uncertainty by allowing flexible input and clear confirmation.

Theme 2: Lack of Self-Location Awareness: Despite successfully identifying destinations, participants consistently struggled to determine their current position on the map while following routes. Although the system supports anchor-based and manual starting location selection, users still found it difficult to maintain awareness of their position during navigation. This lack of continuous self-location feedback increased cognitive load and made navigation more difficult. One participant (P2) stated, *"I see the path, but I don't know where I am on it,"* while another (P4) explained, *"I have to look around and match the map myself. That takes time."* These comments indicate that although the route visualization and manual starting point selection were useful, users still needed clearer and more continuous feedback about their current position to effectively interpret and follow the path

Theme 3: Low Visibility of Floor Transition Cues: Participants had difficulty noticing floor-related interface elements such as floor tabs and cross-floor instructions. Many users failed to recognize when a floor change was required, which led to confusion during navigation tasks. One participant (P1) reported, *"I didn't realize I had to switch floors until you told me."* while another (P4) noted, *"The instruction text is there, but it's easy to miss."* These findings suggest that the visual prominence and clarity of floor transition cues were insufficient, causing users to miss critical navigation steps.

Theme 4: Preference for Passive Navigation Support: Participants expressed a strong preference for passive guidance methods such as audio or turn-by-turn instructions, rather than relying solely on visual map interpretation. One participant (P5) stated, *"I don't want to keep checking my phone while walking,"* while another explained, *"If it told me where to go, I would feel more confident."* A third participant added, *"Audio directions would make this much easier, especially when I'm in a hurry."*

These responses highlight that users prefer navigation systems that minimize interaction effort and allow them to maintain focus on their surroundings. This suggests that passive guidance could significantly improve usability and overall user experience.

Heuristic Evaluation Results: We asked two evaluators to conduct an evaluation using Nielsen's 10 heuristics.

Table 3 Heuristic Evaluation Summary

Heuristic	**Severity (0–4)**	**Finding**
Visibility of system status	1	Floor transition instructions are present but not sufficiently visible, making it difficult for users to notice when a floor change is required.
Match with real world	0	Room numbers and floor labels match real-world environment
User control and freedom	1	Users can clear destinations, but no undo functionality is available

Consistency and standards	1	Floor tabs are consistent across all floors
Error prevention	2	No validation for misspelled room numbers
Recognition rather than recall	1	Destination card reduces memory load
Flexibility and efficiency	1	Recent searches provide some efficiency support, but no advanced shortcuts are available
Aesthetic and minimalist design	1	Interface is generally clean, but the map appears visually cluttered and slightly blurry, which reduces clarity during navigation
Error recovery	2	"No route found" message appears but lacks recovery guidance
Help and documentation	3	No help section or onboarding support is provided

The most severe issue identified was related to Help and Documentation (Severity = 3), indicating that first-time users may struggle to understand the system without guidance. Several moderate issues (Severity = 2), including error prevention and error recovery, suggest that the system lacks sufficient support for handling user mistakes.

The evaluation revealed three key insights:

(i) Search and destination confirmation were efficient, allowing users to quickly locate and verify destinations;
(ii) Users struggled with self-location awareness because the system does not indicate their current position on the map; and
(iii) Floor transition cues were not sufficiently visible, making cross-floor navigation the most difficult and time-consuming task.

What Worked: Several aspects of the system performed well: (i) the search feature was easy to use and enabled quick destination lookup; (ii) the destination card helped users confirm their selection and reduced memory load; (iii) route visualization was effective once users understood their position on the map; and (iv) the interface remained relatively clean and uncluttered.

What Did Not Work: Despite these strengths, several issues were identified: (i) the lack of a “you are here” indicator made it difficult for users to orient themselves; (ii) floor transition cues were easy to miss, causing confusion during navigation; (iii) the absence of audio guidance required users to frequently check the map; and (iv) the system lacked help or onboarding support for first-time users.

Improvements Based on Our Findings: Based on these findings, several improvements are recommended: (i) adding a “you are here” indicator to improve orientation; (ii) making floor transitions more visible or automating them; (iii) incorporating audio or turn-by-turn guidance to support passive navigation; and (iv) providing a simple tutorial or help section for new users.

What We Missed: There are several limitations in our evaluation: (i) the study involved only five participants, limiting generalizability; (ii) it did not consider time-sensitive scenarios; and (iii) features such as voice guidance were not implemented or evaluated.

3. Discussion and Future Work

Learning from this project: This project provided several important insights into designing user-centered navigation systems. We learned that-

I) Identifying user needs early is essential, but not all requirements can be fully implemented due to technical constraints; although users requested a "you are here" indicator during the formative study, the lack of reliable indoor positioning limited full implementation, and we instead used anchor-based and manual starting location selection.

II) Route visualization alone is not sufficient for effective navigation, as users struggled when they could not determine their current position. In addition, we observed that

III) Interface elements that appear obvious to designers may not be noticeable to users, as many participants overlooked floor tabs and transition instructions. Finally, we found that

IV) Users prefer passive guidance such as audio directions or real-time navigation, but these features were not implemented due to time constraints within the course project.

How will we apply this to other projects: We will apply these lessons to future work by ensuring that (i) system constraints are considered early and alternative solutions are designed accordingly, especially when real-time positioning is not available. (ii) We will prioritize early and iterative user testing to validate core interactions before full development. (iii) We will design interfaces with stronger visual hierarchy and clearer feedback so that important elements are easily noticeable. (iv) Finally, we will consider real-world user context and aim to reduce interaction effort during use.

One thing we would do differently: If we were to repeat this project, we would focus on evaluating core usability challenges earlier in the design process. We would test user orientation and self-location issues before full implementation and explore alternative solutions earlier, rather than addressing them after development. This would help reduce rework and improve overall system usability.

4. Participation

Sl	Participant Name	Email	Contribution Areas
1	Tania Khatun	tkhatu1@lsu.edu	Data Collection, Performed Heuristic Evaluation, Wrote Evaluation Design Section
2	FNU Nurjahan	nurja1@lsu.edu	Data Collection, Performed Heuristic Evaluation, Wrote Result section.
3	Nushrat Jahan Ria	nria1@lsu.edu	Performed Evaluation Design, User Testing, Wrote Evaluation design and Result Section.
4	Sidhi Kumari Yadav	SidhiKumari.Yadav@lsu.edu	Performed Evaluation Design, Wrote Discussion and Future Work Section
5	Mingyang Li	myi5@cct.lsu.edu	Data Collection, User Testing, Review Result Evaluation Design Section